

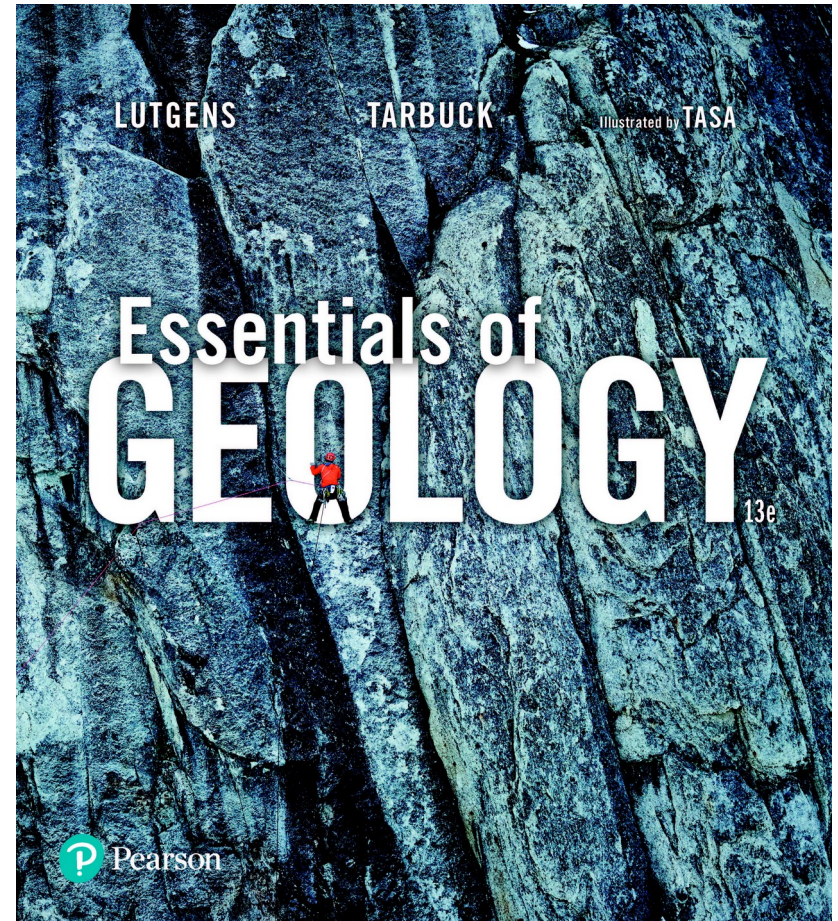
Lecture 7 Outline

Sedimentary Rocks

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Essentials of Geology

Thirteenth Edition



Learning Objectives

- **LO 7.1** Explain the importance of sedimentary rocks and summarize the part of the rock cycle that pertains to sediments and sedimentary rocks. List the three categories of sedimentary rocks.
- **LO 7.2** Describe the primary basis for distinguishing among detrital rocks and discuss how the origin and history of such rocks might be determined.
- **LO 7.3** Explain the processes involved in the formation of chemical sedimentary rocks and list several examples.
- **LO 7.4** Outline the successive stages in the formation of coal.
- **LO 7.5** Describe the processes that convert sediment into sedimentary rock and other changes associated with burial.

Learning Objectives (cont.)

- **LO 7.6** Summarize the criteria used to classify sedimentary rocks.
- **LO 7.7** Distinguish among three broad categories of sedimentary environments and provide an example of each. List several sedimentary structures and explain why these features are useful to geologists.
- **LO 7.8** Distinguish between the two broad groups of nonmetallic mineral resources. Discuss the three important fossil fuels associated with sedimentary rocks.
- **LO 7.9** Relate weathering processes and sedimentary rocks to the carbon cycle.

7.1 An Introduction to Sedimentary Rocks

- Sediments and sedimentary rocks cover approximately 75 percent of Earth.
- They comprise about 5 percent (by volume) of Earth's outer 10 miles.
- Contain evidence of past environments
- Sedimentary rocks are an important resource of
 - Coal, oil, and other fossil fuels
 - Groundwater resources

7.1 An Introduction to Sedimentary Rocks (cont.)

- Sedimentary rocks are products of mechanical and chemical weathering.
 - Sediments and soluble constituents are typically transported downslope by gravity.
 - The sediments are then deposited and subsequently buried.
 - As deposition continues, the sediments are lithified into sedimentary rocks.
- There are three types of sedimentary rocks: Detrital, chemical, and organic sedimentary rocks.

7.2 Detrital Sedimentary Rocks

- Detrital sedimentary rocks form from sediments that have been weathered and transported.
 - The chief constituents of detrital rocks include clay minerals, quartz, feldspars, and micas.
 - Particle size is used to distinguish among the various rock types.

Particle Size Categories

Size Range (millimeters)	Particle Name	Common Name	Detrital Rock
>256	Boulder	Gravel	 
64–256	Cobble		
4–64	Pebble		
2–4	Granule		
1/16–2	Sand	Sand	
1/256–1/16 <1/256	Silt Clay	Mud	 

0 10 20 30 40 50 60 70 mm

7.2 Detrital Sedimentary Rocks

- Shale
 - Silt- and clay-sized particles
 - Form from the gradual settling of sediments in quiet, nonturbulent environments
 - Sediments form in thin layers that are called laminae
 - Has fissility (meaning the rock can be split into thin layers)
 - Crumbles easily and tends to form gentle slopes
 - Most abundant sedimentary rock

Shale



7.2 Detrital Sedimentary Rocks

- Sandstone
 - Sand-sized particles
 - Forms in a variety of environments
 - Second-most abundant sedimentary rock
 - Quartz is the most abundant mineral
 - Quartz sandstone is predominately composed of quartz.
 - Arkose sandstone contains appreciable quantities of feldspar.
 - Graywacke contains rock fragments and matrix, in addition to quartz and sandstone.

Quartz Sandstone



Close up

7.2 Detrital Sedimentary Rocks

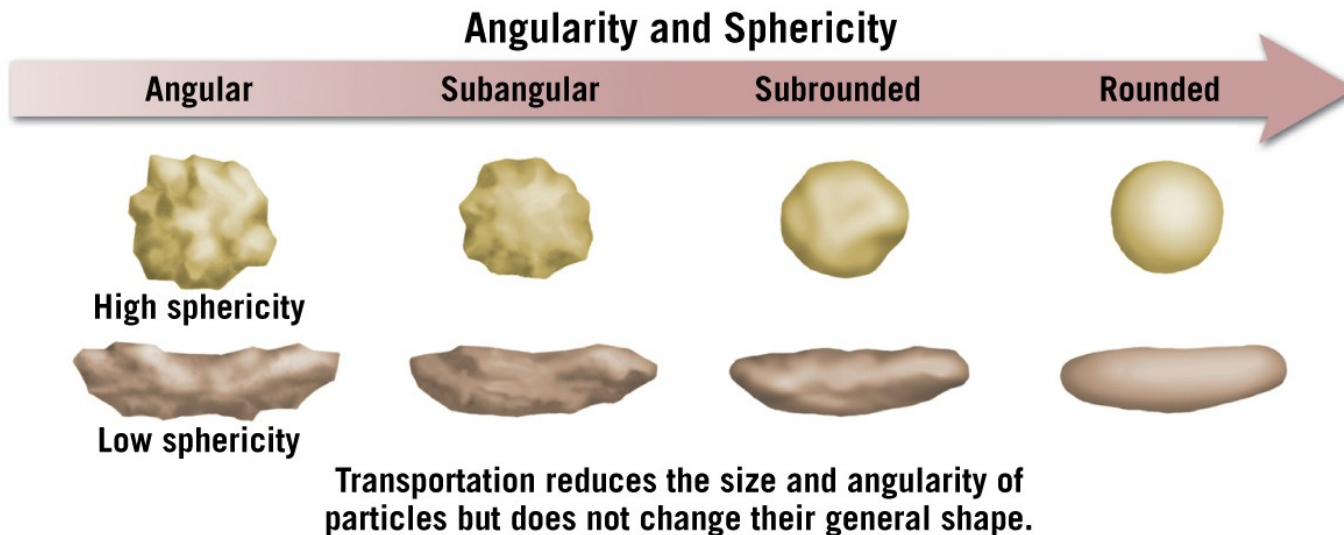
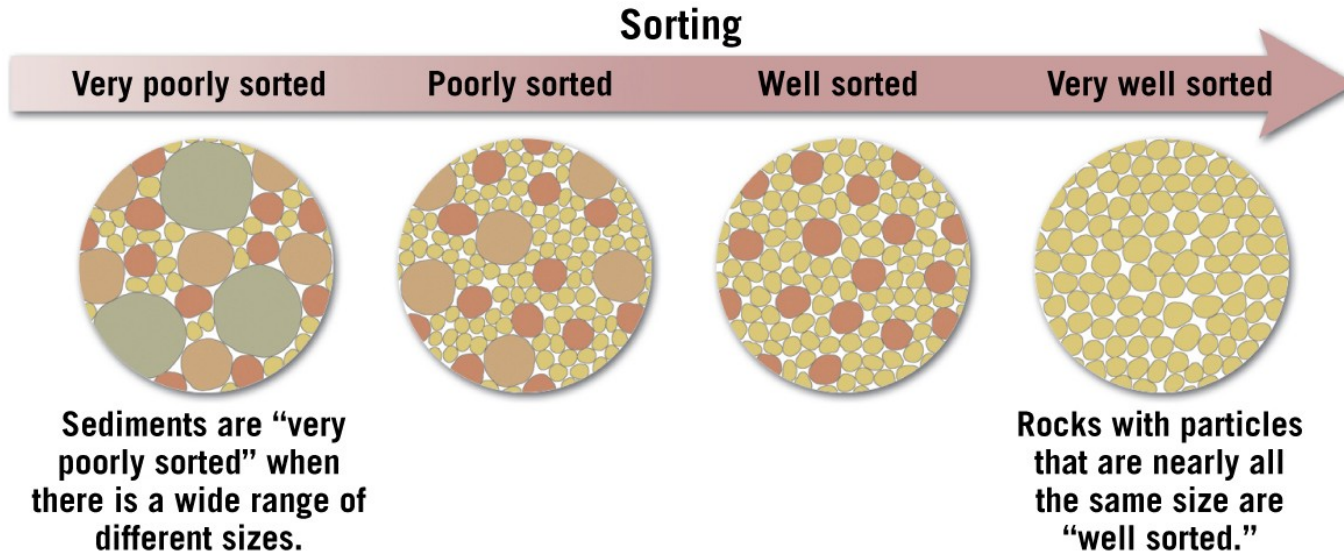
- Sandstone
 - The particles in sandstone vary and are classified by their sorting and shape.
 - Sorting is the degree of similarity in particle size in a sedimentary rock.
 - If all the grains in a rock are of similar size, the rock is well sorted.
 - If the grains in a rock are different sizes (both large and small grains), the rock is poorly sorted.
 - Sorting can help decipher the depositional environment of the rock.

7.2 Detrital Sedimentary Rocks (cont.)

- Sandstone
 - The particles in sandstone vary and are classified by their sorting and shape.
 - Particle shape varies from rounded to angular.
 - The degree of rounding is indicative of how far the sediments have been transported.
 - Rounded sediments are typically transported great distances, while angular sediments are only transported a short distance.

Sorting and Particle Shape

[Tutorial](#)



7.2 Detrital Sedimentary Rocks

- Conglomerate and Breccia
 - Conglomerate consists of rounded, gravel-sized sediments.
 - Breccia consists of angular, gravel-sized sediments.
 - Both types of rocks are poorly sorted.

Conglomerate and Breccia



7.3 Chemical Sedimentary Rocks

- Chemical sedimentary rocks form from precipitated material that was once in solution form.
 - Precipitation of material occurs by:
 - Inorganic processes, such as evaporation
 - Organic processes from water-dwelling organisms form biochemical sedimentary rocks

7.3 Chemical Sedimentary Rocks (cont.)

- Limestone
 - Most abundant chemical sedimentary rock
 - Mainly composed of the mineral calcite
 - Can form from inorganic and biochemical origins

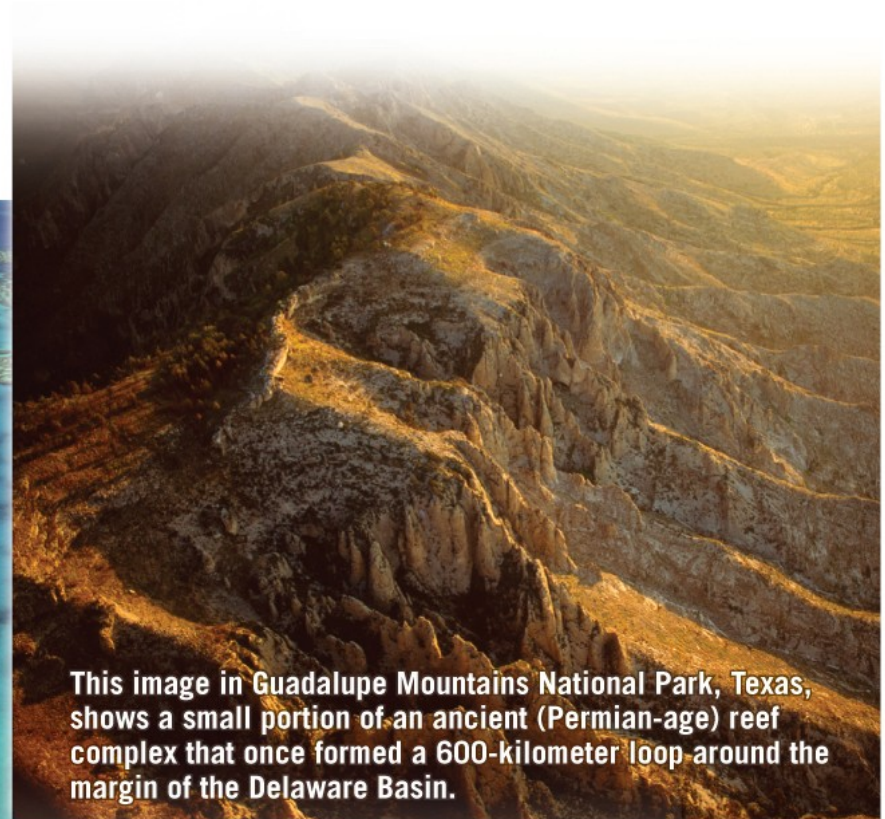
7.3 Chemical Sedimentary Rocks (cont.)

- Limestone
 - Biochemical limestone originates from the shells of marine organisms.
 - Large quantities of marine limestone are formed from corals.
 - Corals secrete a calcium carbonate skeleton and create reefs.
 - Australia's Great Barrier Reef is the largest coral reef on Earth.
 - Coquina is composed of cemented fragments of shell material.
 - Chalk is composed of the tests of microscopic marine organisms.

Carbonate Reefs



Aerial view showing a small portion of Australia's Great Barrier Reef. Located off the coast of Queensland, it extends for 2600 kilometers and consists of more than 2900 individual reefs.



This image in Guadalupe Mountains National Park, Texas, shows a small portion of an ancient (Permian-age) reef complex that once formed a 600-kilometer loop around the margin of the Delaware Basin.

Coquina



Close up

The White Chalk Cliffs

The massive White Chalk Cliffs. Chalk is a biochemical limestone made up almost entirely of the tiny hard parts of microscopic marine organisms, mainly plankton.



View of a group of plankton called *coccolithophores* from a scanning electron microscope. Individual plates shaped like hubcaps are only three one-thousandths of a millimeter in diameter; so tiny they could pass through the eye of a needle.



7.3 Chemical Sedimentary Rocks

- Limestone
 - Inorganic limestone forms when chemical changes increase the calcium carbonate content of the water until it precipitates.
 - Travertine is a type of limestone found in caves.
 - It precipitates when the water in the cave loses carbon dioxide.
 - Oolitic limestone is composed of small spherical grains called ooids.
 - Ooids form as tiny “seeds” roll in shallow marine water supersaturated with calcium carbonate.

Oolitic Limestone



Small spherical grains called ooids are formed by chemical precipitation of calcium carbonate around a tiny nucleus and are the raw material for oolitic limestone.

7.3 Chemical Sedimentary Rocks

- Dolostone
 - Similar to limestone but contains magnesium
 - Origin of dolostone is unclear.
 - Significant quantities of dolostone are created when magnesium-rich waters circulate through limestone.
- Chert
 - Composed of microcrystalline quartz
 - Forms when dissolved silica precipitates
 - Flint, jasper, and agate are varieties of chert

Chert



Flint



Jasper



Chert arrowhead



Petrified wood

7.3 Chemical Sedimentary Rocks

- Evaporites
 - Form when restricted seaways become oversaturated and salt deposition starts
 - Rock salt and rock gypsum are two common evaporites.
 - Occasionally, evaporites form on salt flats when dissolved materials are precipitated as a white crust on the ground.

Salt Flats

Tutorial



This extensive evaporite deposit is a 30,000-acre expanse of hard white salt that in places is nearly 2 meters thick.

7.4 Coal: An Organic Sedimentary Rock

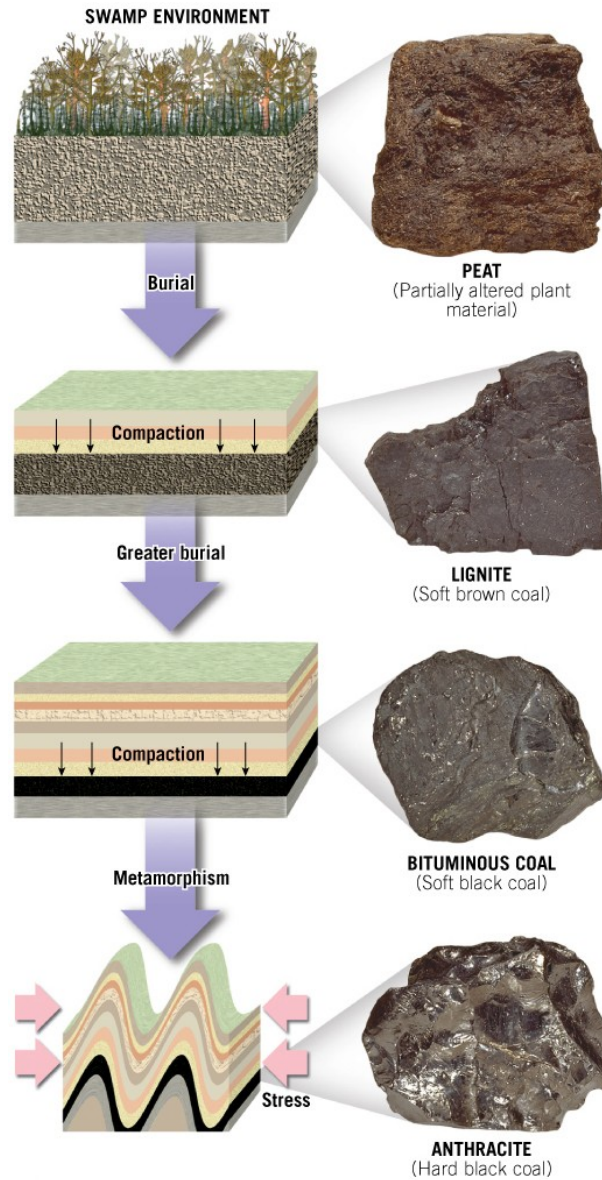
- Organic sedimentary rocks form from the carbon-rich remains of organisms.
- Occasionally, plant structures (leaves, bark, and wood) are identifiable in coal.

7.4 Coal: An Organic Sedimentary Rock (cont.)

- Stages of coal formation
 - Accumulation of plant remains
 - Formation of peat
 - Formation of lignite and bituminous coal
 - Formation of anthracite coal

From Plants to Coal

Tutorial



7.5 Turning Sediment into Sedimentary Rock: Diagenesis & Lithification

- Many changes occur to sediment after it is deposited.
 - Diagenesis: Chemical, physical, and biological changes that take place after sediments are deposited and buried.
 - Occurs within the upper few kilometers of Earth's crust
 - Examples:
 - Recrystallization of more stable minerals from less stable ones (e.g., aragonite to calcite).

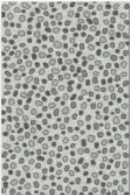
7.5 Turning Sediment into Sedimentary Rock: Diagenesis & Lithification (cont.)

- Many changes occur to sediment after it is deposited.
 - Lithification: Unconsolidated sediments are transformed into sedimentary rocks.
 - Compaction: As sediments are buried, the weight of the overlying material compresses the deeper sediments.
 - Cementation: Involves the crystallization of minerals among the individual sediment grains.

7.6 Classification of Sedimentary Rocks

- Sedimentary rocks are classified according to the type of material.
 - Two major groups
 - Detrital
 - Has clastic texture, composed of discrete fragments cemented together
 - Chemical/Organic
 - Has nonclastic or crystalline texture, where the minerals form patterns of interlocked crystals

Identification of Sedimentary Rocks

Detrital Sedimentary Rocks			
Clastic Texture (particle size)		Sediment Name	Rock Name
Coarse (over 2 mm)		Gravel (Rounded particles)	Conglomerate
		Gravel (Angular particles)	Breccia
Medium (1/16 to 2 mm)		Sand	Sandstone (Arkose)*
Fine (1/16 to 1/256 mm)		Mud	Siltstone
Very fine (less than 1/256 mm)		Mud	Shale or Mudstone

* If abundant feldspar is present the rock is called Arkose.

Chemical and Organic Sedimentary Rocks			
Composition	Texture	Rock Name	
Calcite, CaCO_3	Nonclastic: Fine to coarse crystalline	Crystalline Limestone	
		Travertine	
	Clastic: Visible shells and shell fragments loosely cemented	Coquina	Biochemical limestone
	Clastic: Various size shells and shell fragments cemented with calcite cement	Fossiliferous Limestone	
	Clastic: Microscopic shells and clay	Chalk	
Quartz, SiO_2	Nonclastic: Very fine crystalline	Chert (light colored) Flint (dark colored) Jasper (red)	
Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Nonclastic: Fine to coarse crystalline	Rock Gypsum	
Halite, NaCl	Nonclastic: Fine to coarse crystalline	Rock Salt	
Altered plant fragments	Nonclastic: Fine-grained organic matter	Bituminous Coal	

7.7 Sedimentary Rocks Represent Past Environments

- An environment of deposition or a sedimentary environment is a geographic setting where sediment is accumulating.
- Determines the nature of the sediments that accumulate (grain size, grain shape, etc.).

7.7 Sedimentary Rocks Represent Past Environments (cont.)

- Types of sedimentary environments
 - Three broad categories
 - Continental: Dominated by stream erosion and deposition
 - Marine: Shallow to deep oceanic water
 - Transitional: Shorelines

Sedimentary Environments

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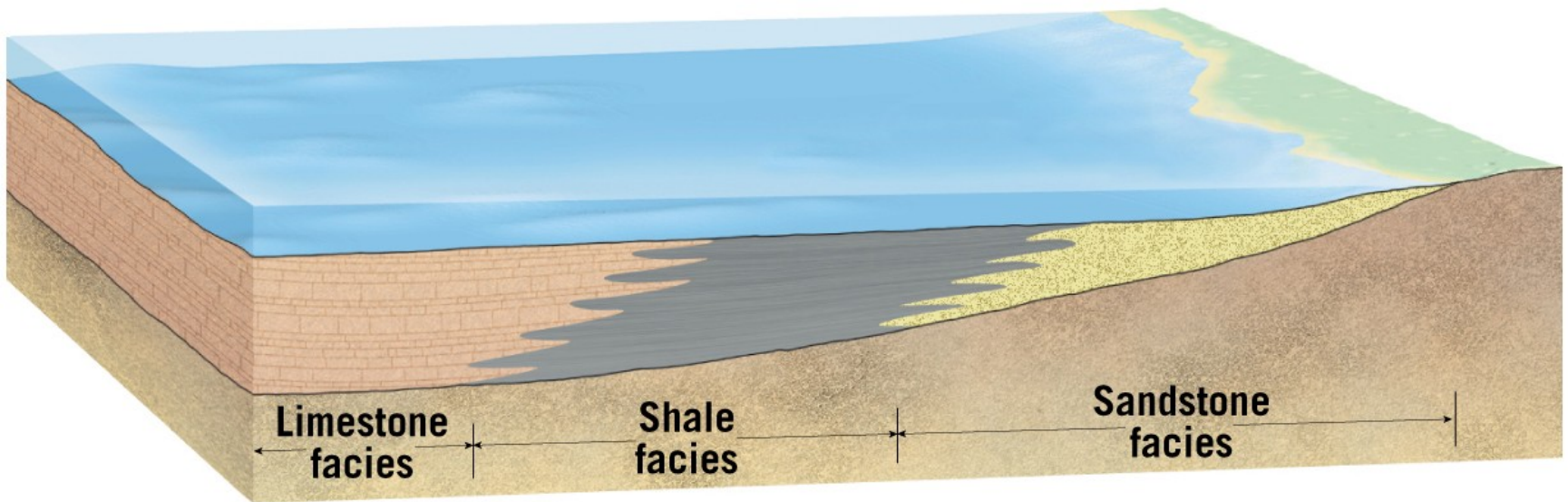


7.7 Sedimentary Rocks Represent Past Environments

- Sedimentary facies
 - Different sediments often accumulate in adjacent environments.
 - Each unit (facies) possesses a distinctive set of characteristics reflecting the conditions of a particular environment.
 - Transitions between different facies are typically gradual.

Sedimentary Facies

Tutorial



7.7 Sedimentary Rocks Represent Past Environments

- Provide additional information useful in the interpretation of Earth's history
- Types of sedimentary structures
 - The layers of the sedimentary rocks are called strata or beds.
 - Most characteristic of sedimentary rocks
 - Bedding planes separate strata.
 - Cross-bedding occurs when the layers in the sedimentary rocks are inclined.
 - Characteristic of sand dunes, deltas, and some stream deposits.

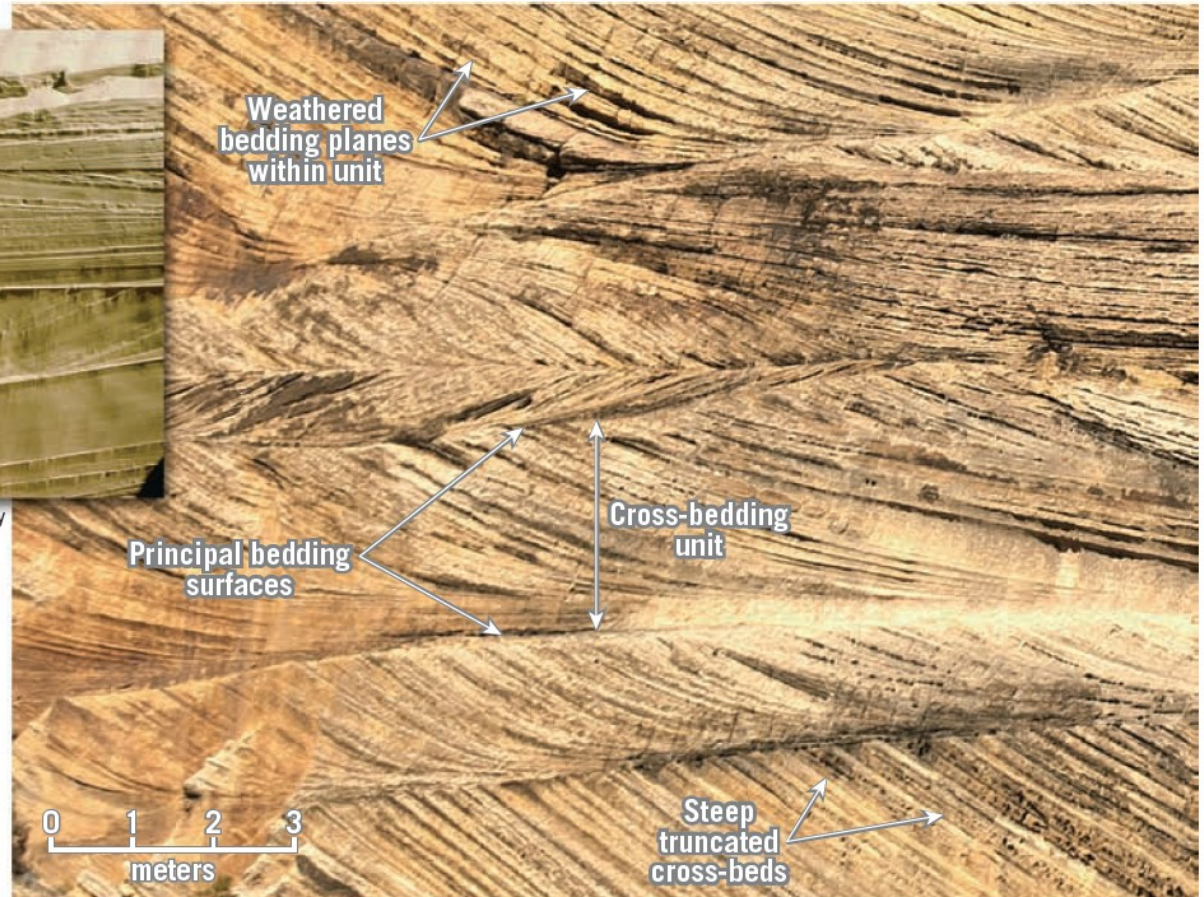
Cross-Bedding



The cutaway section of this sand dune shows the characteristic cross-bedding

John S. Shelton/ University of Washington Libraries

The cross-bedding in this sandstone indicates it was once a sand dune



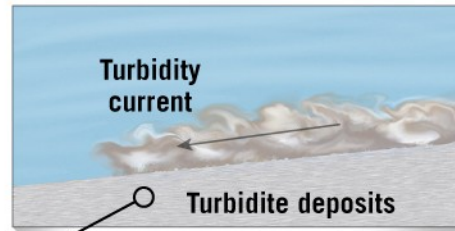
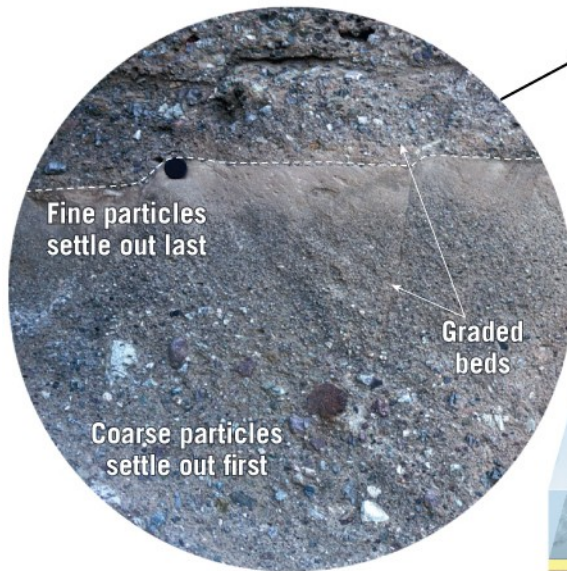
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7.7 Sedimentary Rocks Represent Past Environments

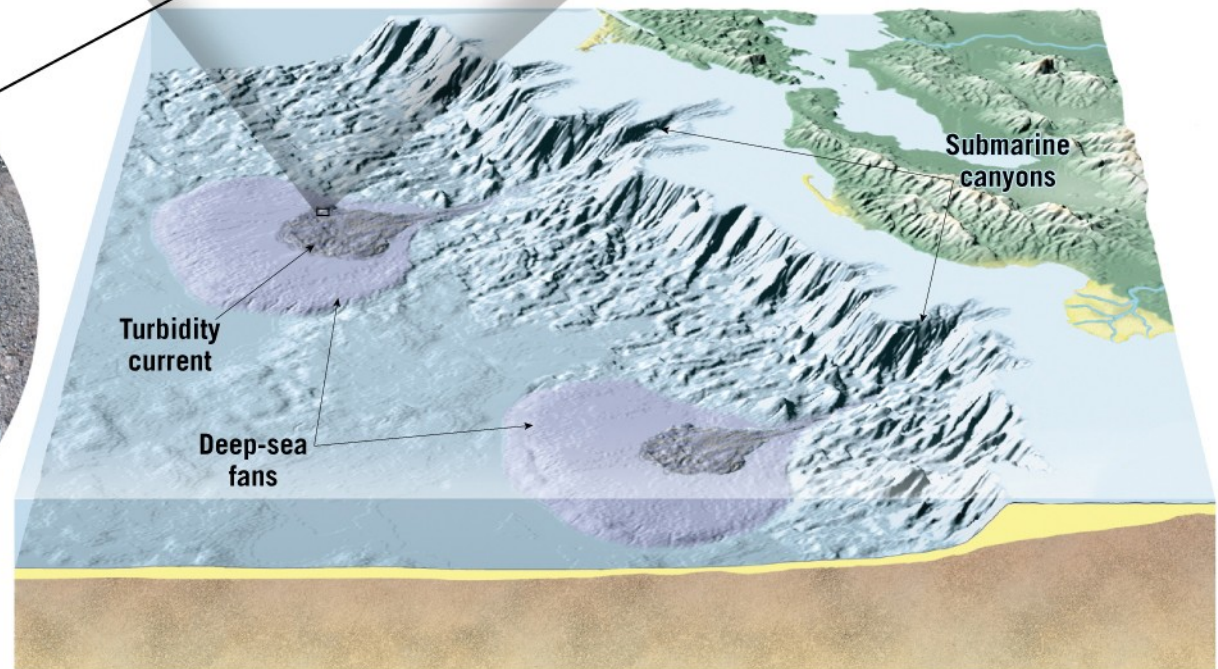
- Types of sedimentary structures
 - Graded beds are a unique situation in which the sediments in a strata gradually change from coarse at the bottom to fine at the top.
 - Often associated with turbidity currents

Graded Bedding

Beds deposited by turbidity currents are called turbidites. Each event produces a single bed characterized by a decrease in sediment size from bottom to top, a feature known as graded bedding.



Turbidity currents are downslope movements of dense, sediment-laden water. They are created when sand and mud on the continental shelf and/or slope are dislodged and thrown into suspension. Because the mud-choked water is denser than normal seawater, it flows downslope, eroding and accumulating more sediment.



7.7 Sedimentary Rocks Represent Past Environments

- Other sedimentary features:
 - Ripple marks are small waves that are lithified in the sedimentary rocks.
 - Mud cracks indicate sediments form in an alternatively wet and dry environment.
 - Fossils are the remains of prehistoric life.

Other Sedimentary Features



A.



B.

7.8 Resources from Sedimentary Rocks

- Nonmetallic mineral resources
 - Use of the word “*mineral*” is broad.
 - Materials not used for fuels or processed for metals are called *nonmetallic mineral resources*.
 - Two common groups
 - Building materials: Crushed stone, gypsum for plaster, clay for bricks
 - Industrial minerals: Fertilizers, salt, sulfur for sulfuric acid

Uses of Nonmetallic Minerals

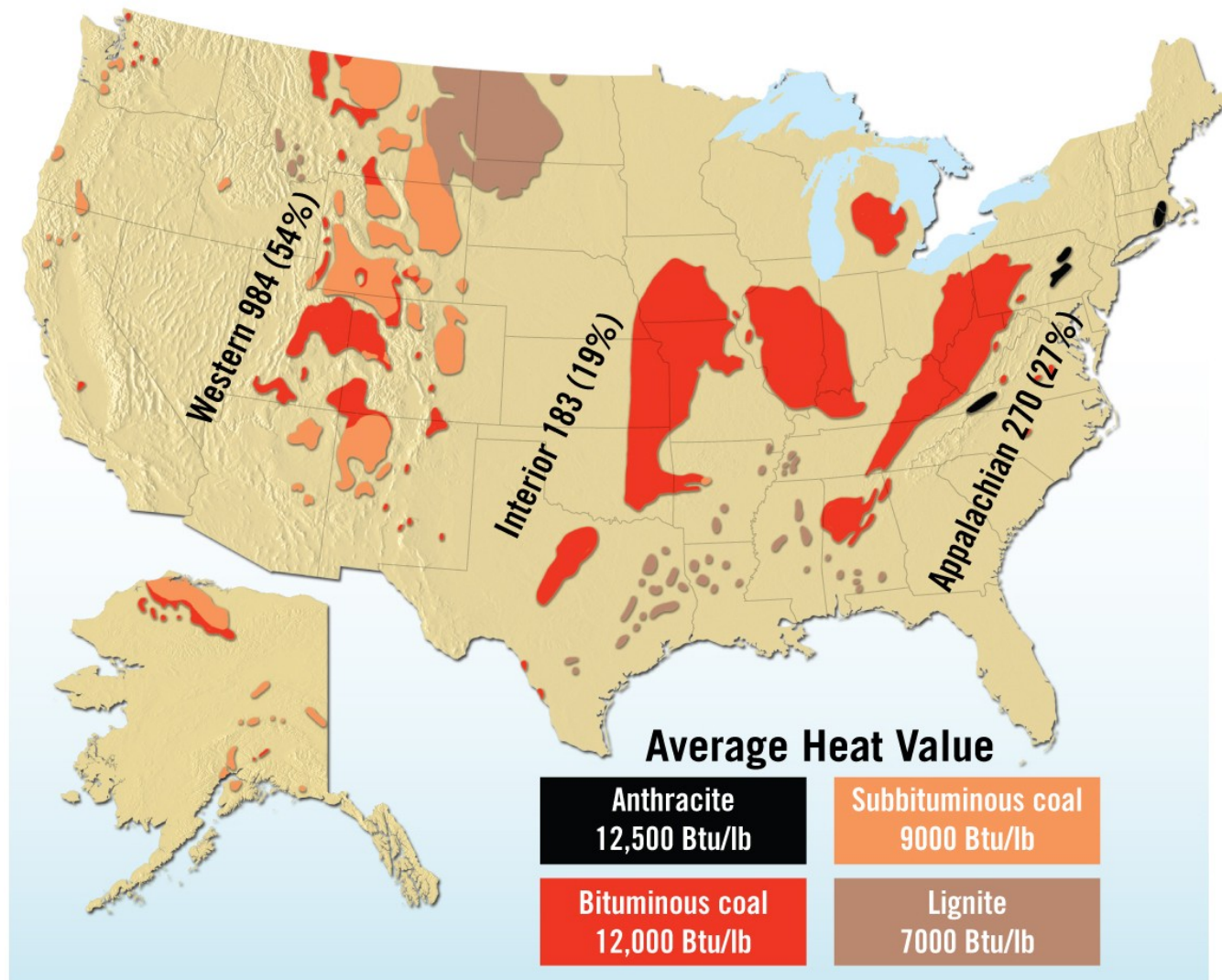
Table 7.1 Uses of Nonmetallic Minerals

Mineral	Uses
Apatite	Phosphorus fertilizers
Asbestos (chrysotile)	Incombustible fibers
Calcite	Aggregate, steelmaking, soil conditioning, chemicals, cement, building stone
Clay minerals (kaolinite)	Ceramics, china
Corundum	Gemstones, abrasives
Diamond	Gemstones, abrasives
Fluorite	Steelmaking, aluminum refining; glass; chemical manufacture
Garnet	Abrasives, gemstones
Graphite	Pencil lead, lubricant, refractories
Gypsum	Plaster of Paris
Halite	Table salt, chemicals, ice control
Muscovite	Insulator in electrical applications
Quartz	Primary ingredient in glass
Sulfur	Chemical and fertilizer manufacture
Sylvite	Potassium fertilizers
Talc	Powder used in paints, cosmetics, etc.

7.8 Resources from Sedimentary Rocks

- Coal
 - Formed mostly from plant material
 - Along with oil and natural gas, coal is commonly called a fossil fuel
 - The major fuel used in power plants to generate electricity
 - Potential environmental problems from mining and air pollution

Coal Fields of the United States



7.8 Energy Resources from Sedimentary Rocks

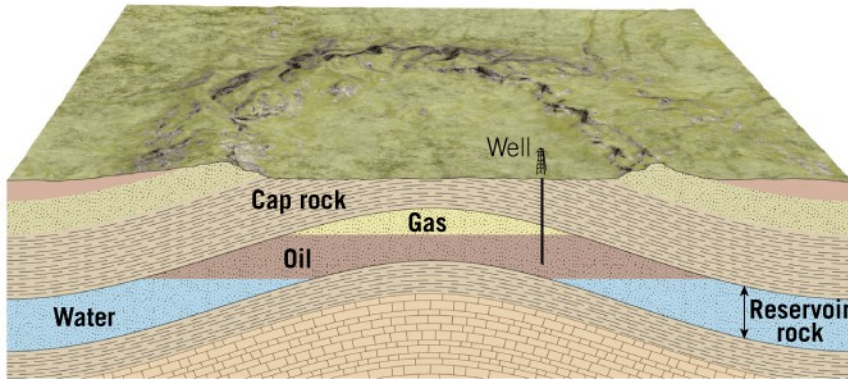
- Oil and natural gas
 - Derived from the remains of marine plants and animals
 - Both are composed of various hydrocarbon compounds and found in similar environments.
 - Oil trap: Geologic environment that allows significant amounts of oil and gas to accumulate.

7.8 Energy Resources from Sedimentary Rocks (cont.)

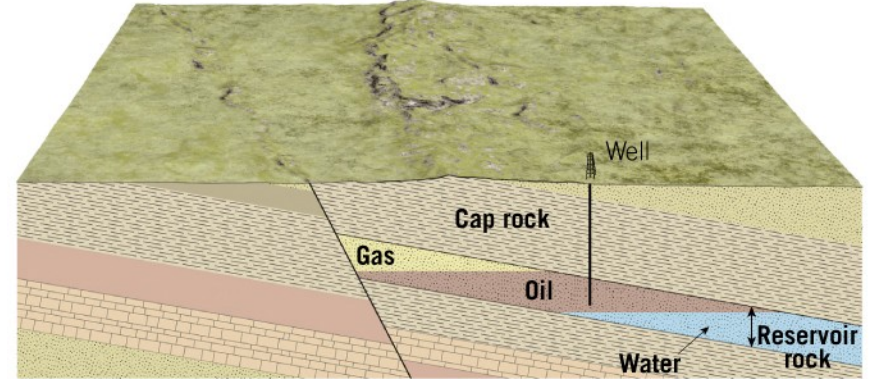
- Oil and natural gas
 - Two basic conditions for an oil trap
 - Porous, permeable reservoir rock
 - Impermeable cap rock, such as shale
 - Cap rock keeps the mobile oil and gas from escaping at the surface.
 - Hydraulic fracturing (shattering rocks underground) may be used to release gas and oil from rocks with low permeability, like shale.

Common Oil Traps

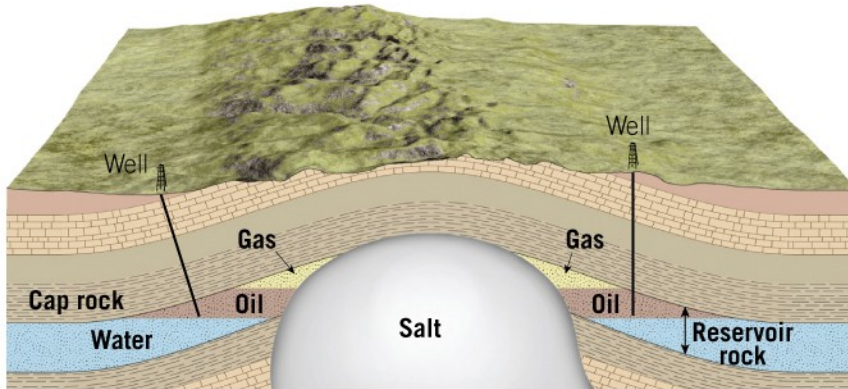
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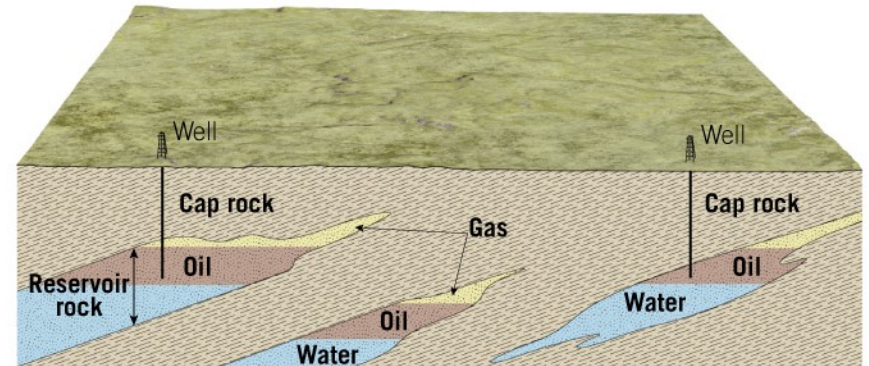
A. Anticline



B. Fault trap

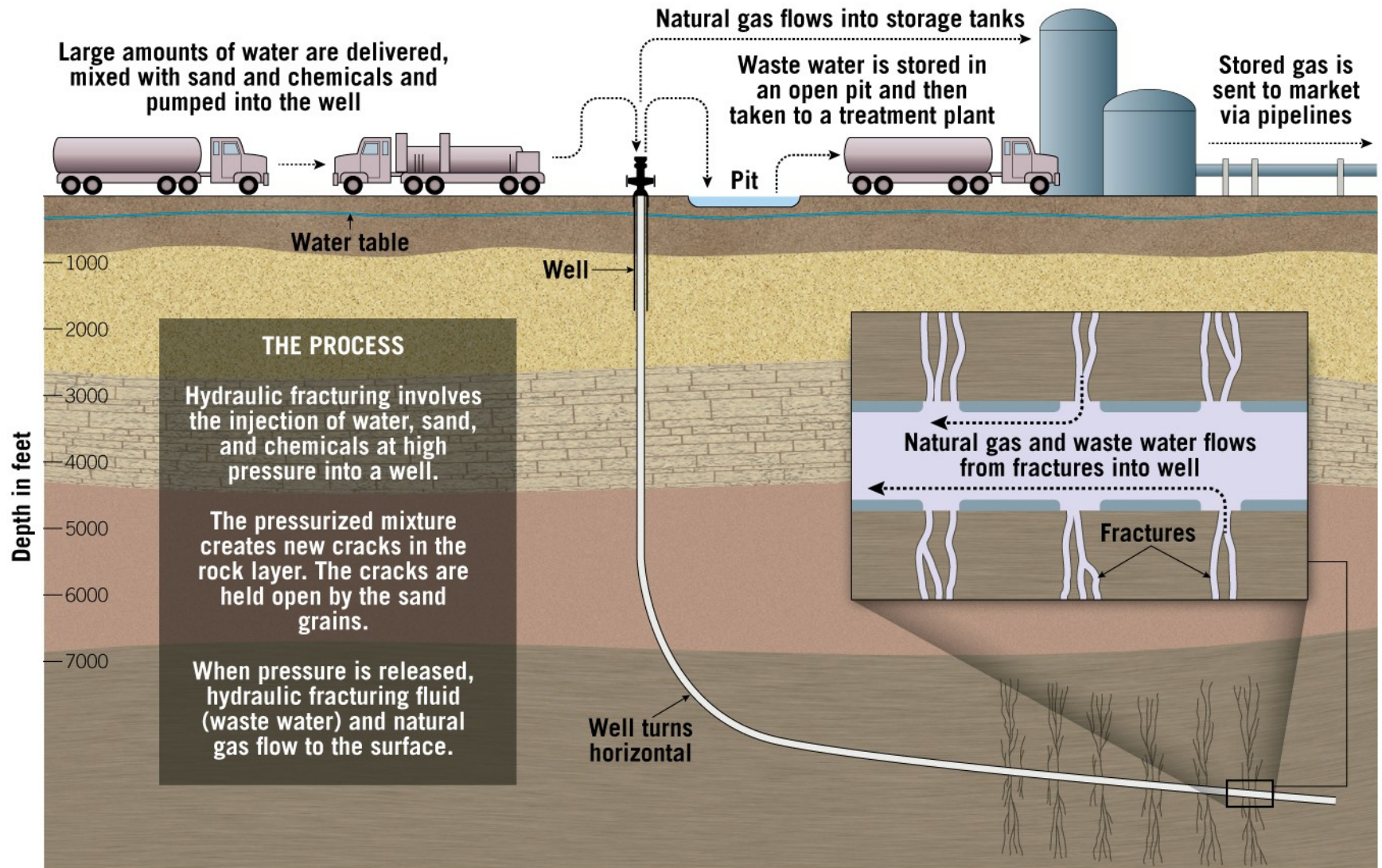


C. Salt dome



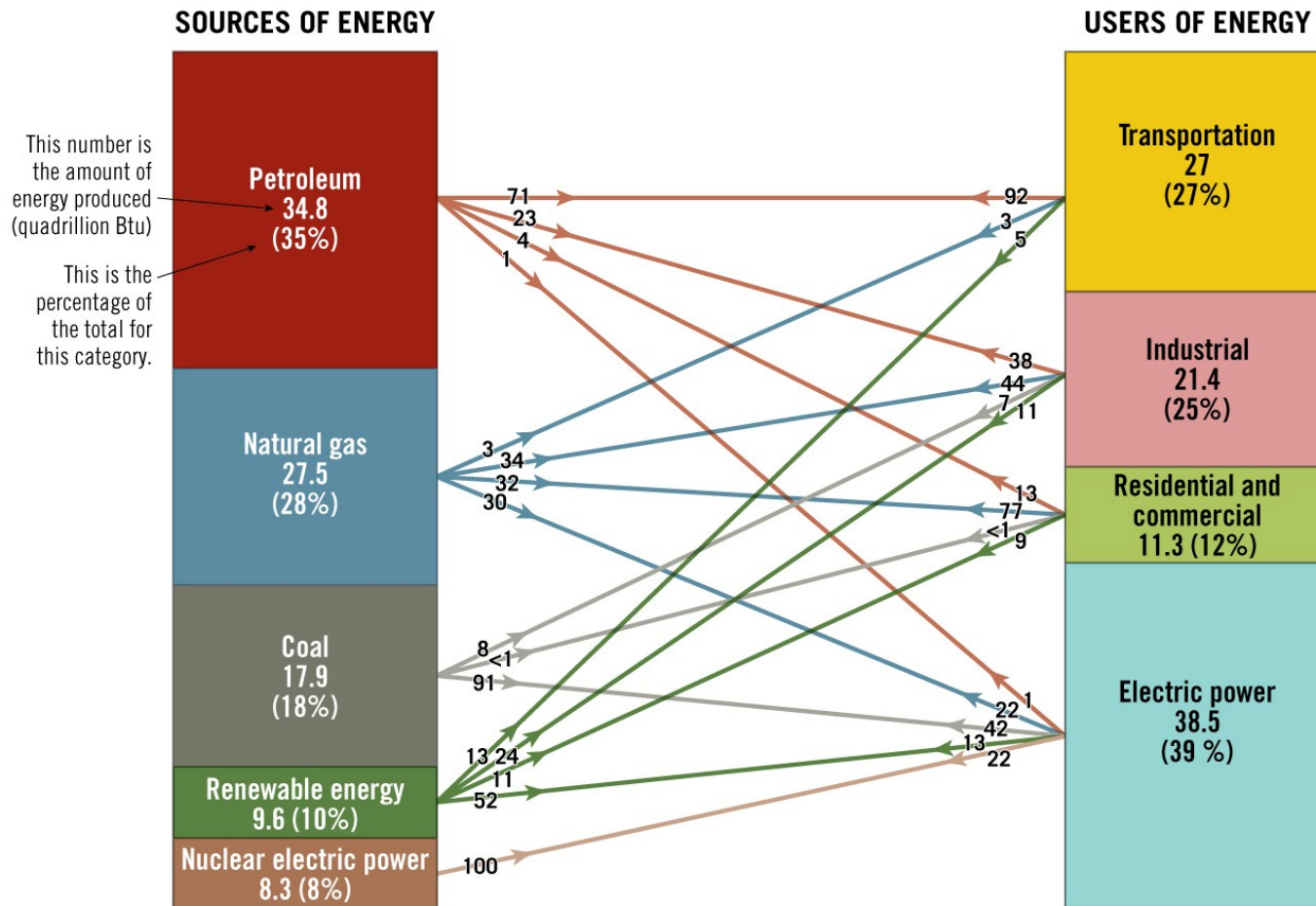
D. Stratigraphic (pinchout) trap

Hydraulic Fracturing (“Fracking”)



U.S. Energy Consumption, 2014

Tutorial



Reading this double graph:

The left side indicates what energy sources we use. The right side shows where we use the energy. The lines with numbers that connect the graphs provide more details. Use the top line as an example. It shows that 71% of the petroleum is used by the transportation sector. It also indicates that 92% of the energy used by the transportation sector is petroleum.

7.9 The Carbon Cycle & Sedimentary Rocks

- Carbon dioxide (CO₂) is one of the most active parts of the carbon cycle.
 - Atmosphere: CO₂ is a greenhouse gas
 - Biosphere: Photosynthesis by vegetation
 - Over geologic time, considerable amounts of plant biomass are converted into fossil fuels in sediments.
 - When fossil fuels are burned, CO₂ is released back into the atmosphere.
- Carbon is also found in the hydrosphere and geosphere.

Carbon Cycle

